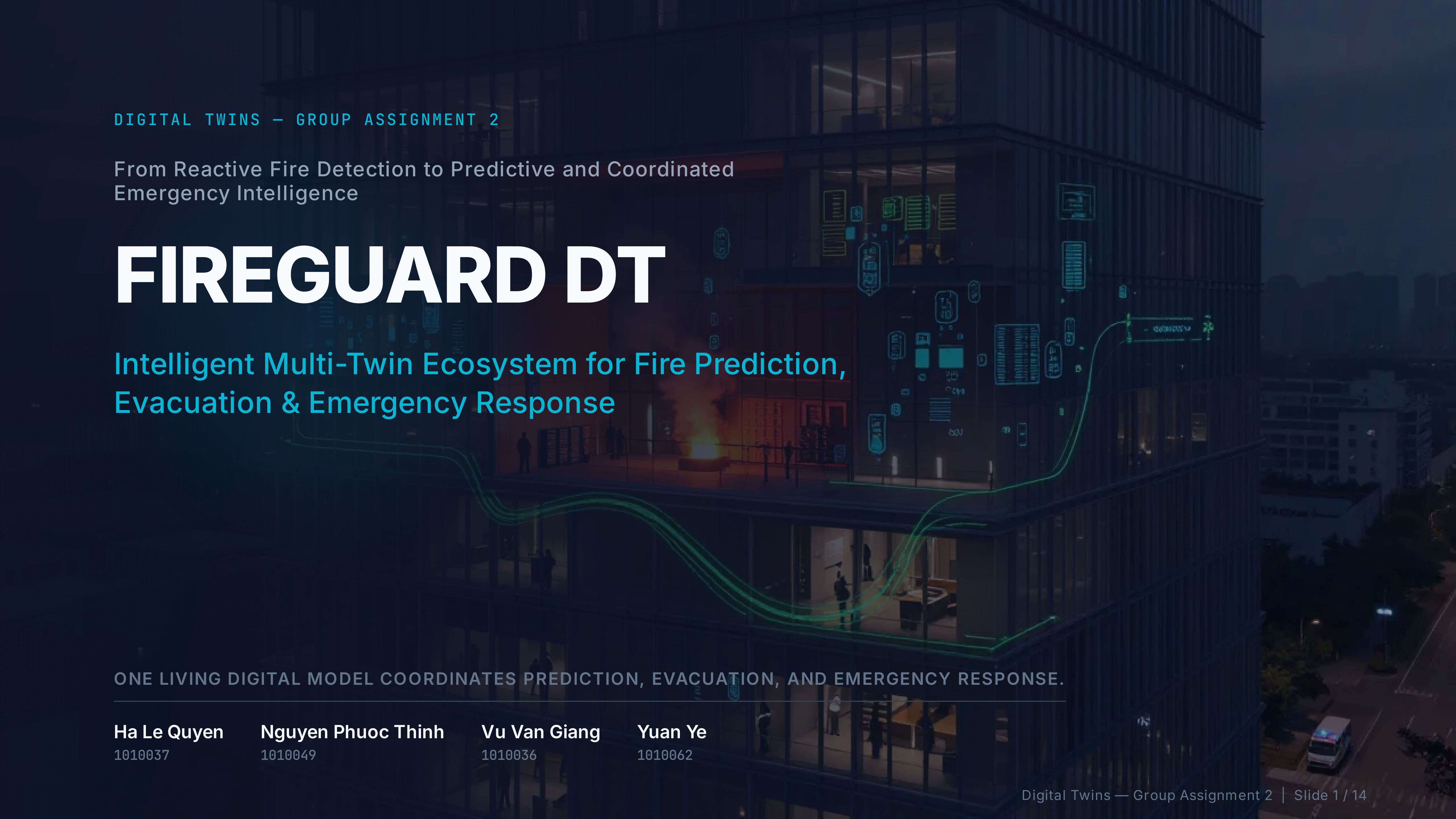

DIGITAL TWINS — GROUP ASSIGNMENT 2

From Reactive Fire Detection to Predictive and Coordinated Emergency Intelligence

FIREGUARD DT

Intelligent Multi-Twin Ecosystem for Fire Prediction, Evacuation & Emergency Response

ONE LIVING DIGITAL MODEL COORDINATES PREDICTION, EVACUATION, AND EMERGENCY RESPONSE.

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THE PROBLEM: TRADITIONAL FIRE RESPONSE IS REACTIVE

DIGITAL TWIN GROUP ASSIGNMENT 2 // INTELLIGENT ECOSYSTEM

Fire emergencies evolve faster than static safety plans.

TRADITIONAL	Detect	Alarm	Evacuate	Respond		
FIREGUARD DT	Monitor	Predict	Simulate	Optimize	Coordinate	Respond

Late Awareness

- Alarms react **after** abnormal conditions are detected.
- Teams receive limited forward-looking insight into risk.

Static Response

- Predefined evacuation routes are rigid and non-adaptive.
- Failure to account for smoke, congestion, or blocked exits.

Fragmented Intelligence

- Building, occupancy, and environmental data remain isolated.
- Lack of unified cross-domain situational awareness.

⚠ Trust Challenge: Unexplained AI predictions create governance and accountability concerns.

Can a Digital Twin predict emerging risk and continuously coordinate the safest response before conditions become critical?

FIREGUARD DT — INTELLIGENT MULTI-TWIN ECOSYSTEM

4 INTERACTING DIGITAL TWINS + AI DECISION ORCHESTRATOR + SHARED 3D SPATIAL MODEL

Fire & Environment Twin

- Predicts emerging fire risk profiles
- Tracks Temp, Smoke, CO/CO_2, Humidity
- Monitors electrical load & HVAC effects

Building Infrastructure Twin

- Rooms, corridors, and exit status
- HVAC zones & sprinkler health
- Power grid & critical asset monitoring

CENTRAL INTELLIGENCE HUB

AI / DECISION ORCHESTRATOR

State Fusion	Prediction
Safety Rules	Optimization
Explainability	Oversight

Cross-twin decision layer
Fuses state, predictions, safety rules, optimization, and human oversight across the four domain twins.

Occupancy & Evacuation Twin

- Real-time people distribution mapping
- Congestion & vulnerability assessment
- Calculates dynamic safe evacuation routes

Emergency Response Twin

- Coordinates fire crews & drones
- Identifies optimal access points
- Resource allocation & dispatch management

4 domain twins + AI orchestration + shared 3D spatial context — not a fifth twin > one coordinated response

INTEGRATED MULTI-TWIN ECOSYSTEM

DIGITAL TWIN GROUP ASSIGNMENT 2 // INTELLIGENT ECOSYSTEM

AI / DECISION ORCHESTRATOR

Prediction Engine | Risk Fusion | Safety Rules | Path Optimization | Decision Log

EVENT SYNCHRONIZATION

Fire & Environment
Sensors, HVAC, Smoke propagation models

Building Infrastructure
Static BIM, Assets, Structural integrity

Occupancy & Evacuation
People flow, Vulnerability, Dynamic routes

Live Sensor Events · Twin-State Updates · Risk Alerts · Confirmed Actions

4TH DOMAIN TWIN — EMERGENCY RESPONSE

Coordinates fire crews, inspection drones, access points, and resource dispatch.

COMMON SPATIAL CONTEXT — NOT A DIGITAL TWIN

Shared 3D Building Spatial Model

All four twins reference the same spatial model for hazards · infrastructure · people · routes · response resources.

Coordination Strategy

HYBRID / FEDERATED ARCHITECTURE

Domain Autonomy

Each twin maintains its domain-specific state and specialized simulation models independently.

Event + Spatial Backbone

Twins exchange events through a shared backbone and interpret them against one common 3D building context.

Central Orchestration

A centralized AI layer fuses multi-twin states to coordinate complex, cross-domain decisions.

Safety Governance

Critical actions are strictly filtered through deterministic safety rules and human approval gates.

Assignment Evidence: four logical twins + mapped event flow + shared 3D spatial context + governed coordination.

3D EMERGENCY SCENARIO: FROM PREDICTION TO ACTION

SHARED 3D SPATIAL MODEL CONNECTS HAZARDS, INFRASTRUCTURE, PEOPLE, ROUTES, AND RESPONSE RESOURCES

01

PREDICT

Electrical-room sensors:

Temp ↑, Smoke ↑, CO ↑

ML RISK → CRITICAL

Critical Recall: 98.5%

02

SPATIALIZE

Shared 3D Model maps:

Elec. Room → Corridor C → HVAC Zone 3

Places the hazard and smoke-propagation path inside one common building context.

03

LOCATE PEOPLE

Occupancy Twin locates people within affected 3D zones

43
Occupants at Risk

04

OPTIMIZE

Smoke makes Corridor C unsafe.

Route A: Unsafe

Route B: Recommended

Dynamic recalculation based on current hazard state.

05

RESPOND

Response Twin:

Dispatch drone & crew

42 sec Containment

2 Resources

06

COORDINATE

Orchestrator:

Validate safety rules

Request human approval

Coordinate twin actions

Log decision trail.

HVAC Isolation: Approved

SHARED 3D BUILDING MODEL

COMMON CONTEXT — NOT A FIFTH TWIN

Electrical Room → Smoke / Corridor C → 43 occupants → Safe Route B → Crew access

The shared 3D model turns distributed twin states into one spatially understandable emergency picture.

PREDICTIVE INTELLIGENCE

DIGITAL TWIN GROUP ASSIGNMENT 2 // INTELLIGENT ECOSYSTEM

01 // DATA INPUTS

- Temperature
- Smoke Conc.
- CO / CO_2
- Humidity
- Electrical Load
- Occupancy
- HVAC State
- Sequence Data

02 // MODEL PIPELINE

- Logistic Regression (Baseline)
- Random Forest Classifier
- Gradient Boosting / XGBoost

EVALUATION METRICS

Accuracy	97.9%
Precision	97.2%
Recall	97.6%
F1-Score	97.4%
ROC-AUC	99.8%
Critical Recall	98.5%

03 // ACTIONABLE OUTPUT

NORMAL	10.1%
WARNING	89.9%
CRITICAL	0.0%

Source: ML_MODEL
Model:fire-risk-v1

Evaluation Principle:
Prioritize **Recall** for dangerous conditions. A missed hazard is more costly than a false warning.

EXPLAINABLE & TRUSTWORTHY AI

DIGITAL TWIN GROUP ASSIGNMENT 2 // INTELLIGENT ECOSYSTEM

CURRENT PREDICTION **Warning**

89.9% Confidence

Live inference from Fire & Environment Twin

EXPLAINABILITY METHODS

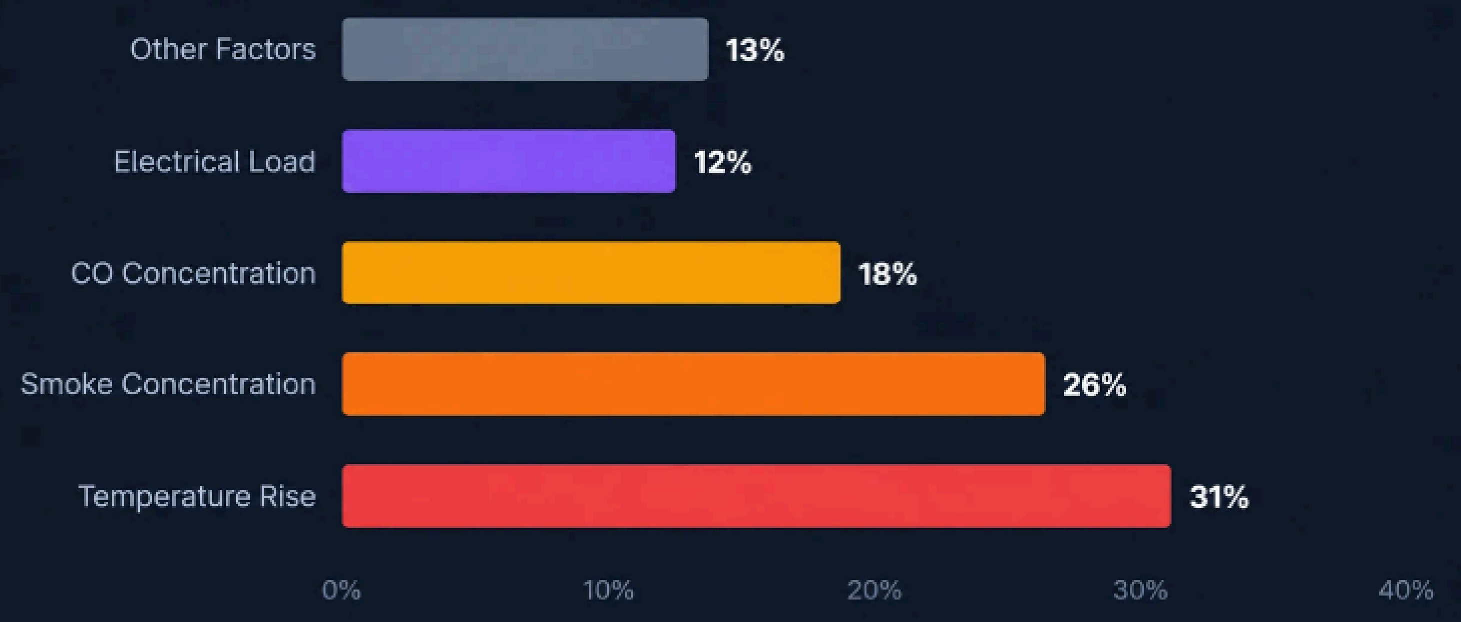
- Global feature importance
- Local logistic contribution analysis
- Prediction confidence
- Physical-consistency check against sensor patterns

CONTROLLED-AUTONOMY LADDER

AI Prediction → Safety Rules → Confidence Check → Human Approval → Critical Action

OPERATOR QUESTIONS ANSWERED

- Why was this prediction generated?
- Which signals contributed most?
- Is the prediction consistent with physical conditions?



AI supports *emergency decisions — it does not blindly replace human responsibility.*

DYNAMIC EVACUATION OPTIMIZATION

DIGITAL TWIN GROUP ASSIGNMENT 2 // INTELLIGENT ECOSYSTEM

TRADITIONAL APPROACH

Static paths optimized for distance. Fails when routes are blocked by smoke or fire.

FIREGUARD DT

Dynamic paths optimized for safety. Adapts to real-time environmental hazards.

$$C(P) = w_1 D(P) + w_2 R_{fire}(P) + w_3 R_{smoke}(P) + w_4 C_{crowd}(P)$$

$D(P)$: Distance

$R_{fire}(P)$: Fire Risk

$R_{smoke}(P)$: Smoke Risk

$C_{crowd}(P)$: Congestion

$$P* = \arg\min_P C(P)$$

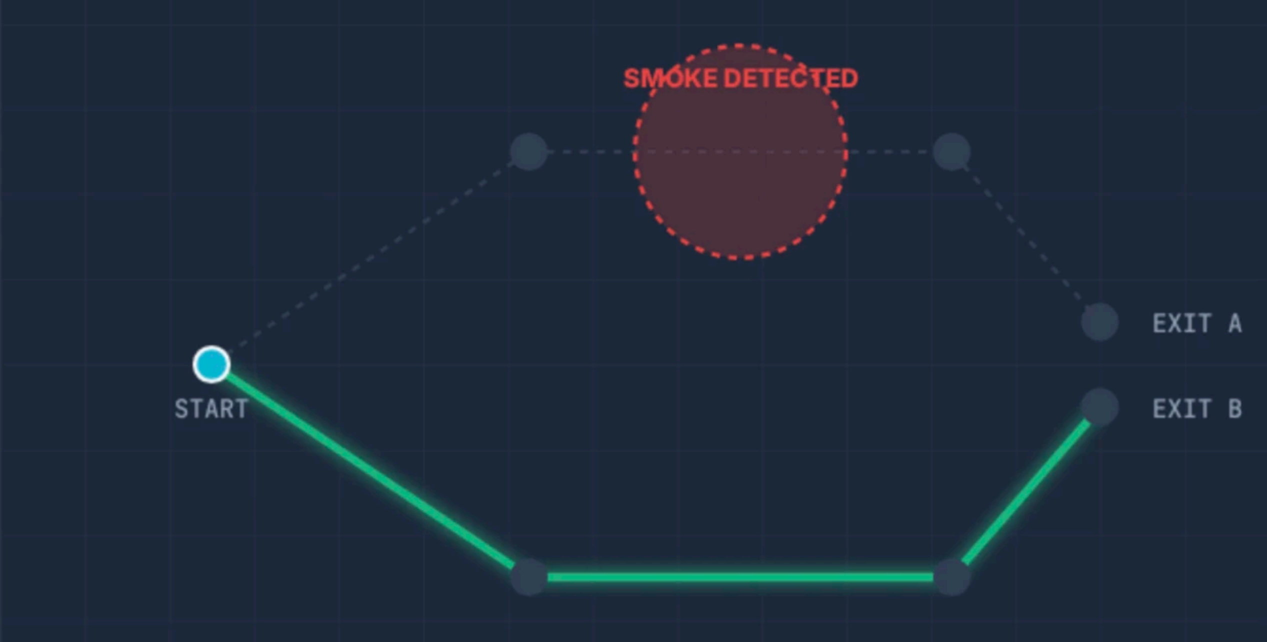
- Algorithm: Dynamic Dijkstra
- Constraints: Blocked zones = ∞ cost
- Priority: Safety & hazard exposure

EXECUTIVE TAKEAWAY

Shortest routes are not always the safest. FireGuard DT ensures the safest feasible path is prioritized.

DYNAMIC ROUTE RECALCULATION

REPLAY SIMULATION



ROUTE: SAFEST

3D WHAT-IF SCENARIO LAB

ONE SHARED 3D BUILDING CONTEXT FOR HAZARDS, SMOKE, PEOPLE, ROUTES, AND RESPONSE RESOURCES

3D Scenario Configuration

COMMON SPATIAL CONTEXT — NOT A FIFTH TWIN

FIRE ORIGIN Electrical Room	FIRE SEVERITY High
OCCUPANCY 250 Persons	EXIT B STATUS Blocked
HVAC STATE Running	SPRINKLER Failed

STRATEGY	EVACUATION TIME	HAZARD EXPOSURE	PEAK CONGESTION
Twin Optimized	55 s	19.319	0.875

Scenario Library

Normal Response	Blocked Exit
Peak Occupancy	Smoke Propagation
Sprinkler Failure	Sensor Malfunction

Spatial Simulation Outputs

- Render fire origin, smoke/HVAC propagation, blocked areas, and infrastructure state in the shared building model.
- Recalculate people movement, safe routes, congestion, and crew access against the same spatial conditions.
- Compare evacuation time, dangerous-zone exposure, peak congestion, and response time across strategies.

Key Message: The shared 3D spatial model lets the team test coordinated safety strategies before applying them to the physical environment.

CYBERSECURITY & RESILIENCE

DIGITAL TWIN GROUP ASSIGNMENT 2 // INTELLIGENT ECOSYSTEM

Device & Communication

- Device identity & authentication
- TLS-encrypted communication
- Signed / secured APIs

Platform Security

- Role-Based Access Control
- Network segmentation
- Audit logging
- Encryption at rest

Sensor Trust

- Multi-sensor consistency checks
- Plausibility / range checks
- Anomaly detection
- Source & timestamp validation

Resilience

- Edge safety rules
- Local alarm continuity
- Graceful degradation
- Recovery & sync after reconnection

ILLUSTRATIVE ANOMALY DETECTION SEQUENCE

```
Sensor A — Temperature: 95°C ↑
Sensor B — Temperature: 26°C ✓
Smoke: Normal  ·  CO: Normal
→ Potential sensor anomaly detected
→ Request validation — do not auto-declare fire from one inconsistent sensor
```

THREAT-TO-CONTROL MAPPING

Spoofed sensor	→ Multi-sensor consistency checks
Compromised account	→ RBAC + MFA enforcement
Network outage	→ Edge fallback & local continuity
Tampered decision	→ Signed immutable audit trail

GOVERNANCE, ETHICS & RESPONSIBLE AUTONOMY

DIGITAL TWIN GROUP ASSIGNMENT 2 // INTELLIGENT ECOSYSTEM

Transparency

- Explainable predictions (SHAP)
- Confidence level reporting
- Immutable decision audit trails

Privacy

- Anonymous occupancy counting
- PII-free data minimization
- Secure zone-based isolation

AI Safety

- Mandatory human-in-the-loop
- Deterministic rule validation
- Fail-safe edge operation

Bias & Reliability

- Zone-specific model evaluation
- Vulnerable occupant priority
- Cross-scenario reliability tests

RISK-BASED AUTONOMY MATRIX

DECISION TYPE	RISK LEVEL	RECOMMENDED CONTROL MECHANISM
System Health & Dashboard Updates	LOW	Fully Autonomous background processing
Evacuation Route Recalculation	MEDIUM	Autonomous with deterministic rule-safety validation
HVAC / Smoke-Control Mode Change	HIGH	Strong validation + Mandatory authorized approval
Critical Resource Dispatch (Crews/Drones)	HIGH	Human-in-the-loop + Multi-stakeholder audit trail

Governance Principle: *Automation level is strictly proportional to decision risk. AI supports emergency decisions; it does not replace human responsibility.*

BUSINESS VALUE AND ROI

DIGITAL TWIN GROUP ASSIGNMENT 2 // INTELLIGENT ECOSYSTEM

Safety Benefits

- Earlier risk identification through predictive analytics
- Faster emergency response with coordinated twins
- Safer evacuation via dynamic route optimization

Operational Efficiency

- Reduced false alarms and unnecessary evacuations
- Improved emergency resource utilization.
- Accelerated incident investigation and reporting

Financial Protection

- Reduced business disruption and downtime
- Potential property damage and insurance reduction
- Optimized resource utilization during emergencies

ROI Methodology

```
ROI = (Total Benefits - Total Costs) / Total Costs × 100%
```

INVESTMENT COSTS

IoT Sensor Integration
Twin Platform Licensing
ML Model Development
Security Hardening
Staff Training & Drills

PROJECTED BENEFITS

Avoided Downtime
Damage Risk Reduction
Maintenance Savings
Response Efficiency
Risk Compliance Value

Scenario Analysis

Metric	Conservative	Base	Optimistic
Initial Investment	880K SGD	820K SGD	820K SGD
Annual Benefit	505K SGD	820K SGD	1120K SGD
Payback Period	35.2 months	15.5 months	10.41 months
3-Year ROI	1.34%	78.91%	149.81%

Financial values are illustrative project assumptions for academic feasibility analysis, not observed production financial results.

STRATEGIC DEPLOYMENT ROADMAP

DIGITAL TWIN GROUP ASSIGNMENT 2 // INTELLIGENT ECOSYSTEM

PHASE 01
Digital Pilot

- 1 Floor deployment
- Fire + Building Twins
- Simulated sensor nodes
- Initial ML baseline

PHASE 02
Intelligent Pilot

- Full Building scope
- Occupancy Twin integration
- Dynamic evacuation engine
- Explainable AI module

PHASE 03
Operational

- Response Twin activation
- Full IoT sensor network
- Edge intelligence nodes
- Security hardening

PHASE 04
Campus Ecosystem

- Multi-building expansion
- Shared response resources
- Central Command Center
- Cross-building sync

PHASE 05
Intelligent Ecosystem

- Continuous ML monitoring
- Predictive maintenance
- Resource optimization
- Controlled autonomy

VALIDATION STAGE GATES

Model
Target recall/F1 and false-alarm tolerance

Safety
Successful scenario tests & rule verification

Security
Pen-testing & incident recovery exercises

Operations
Staff training & emergency-drill acceptance

Business
Approved ROI assumptions & pilot evidence

Scaling from focused pilot to autonomous campus-wide intelligence.

RESULTS AND CONCLUSION

DIGITAL TWIN GROUP ASSIGNMENT 2 // INTELLIGENT ECOSYSTEM

FIRE & ENV TWIN

INFRASTRUCTURE TWIN

AI / DECISION ORCHESTRATOR

RESPONSE TWIN

OCCUPANCY TWIN

PREDICT
Identify emerging fire risks early.

UNDERSTAND
Unified view of building context.

OPTIMIZE
Safest dynamic evacuation paths.

COORDINATE
Sync resources and response.

GOVERN
Trust through explainability.

FireGuard DT transforms a fire alarm from a reactive warning into a predictive, coordinated, and adaptive emergency-response ecosystem.

PERFORMANCE METRICS

System Impact

PREDICTION MACRO (F1)

97.4%

TWIN-OPTIMIZED EVACUATION

55 s Scenario simulation result

FIRST RESPONSE TIME

37 s Emergency simulation result

PROJECTED 3-YEAR ROI

78.91%* Illustrative base scenario

Note: Technical metrics are computed from synthetic test data and deterministic scenario simulations. ROI is an illustrative academic feasibility estimate based on explicit project assumptions.